

Single-cell RNA-sequencing

- Why single-cell ?

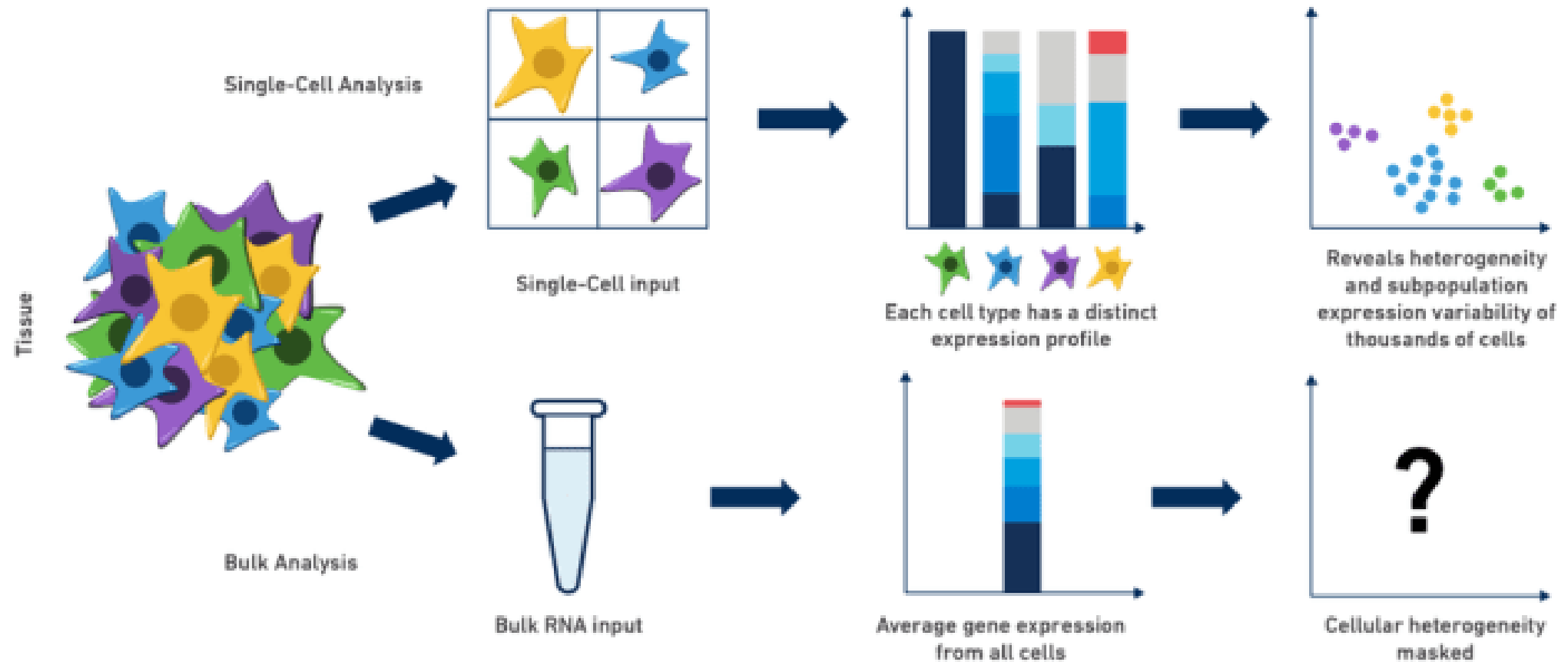


Bulk

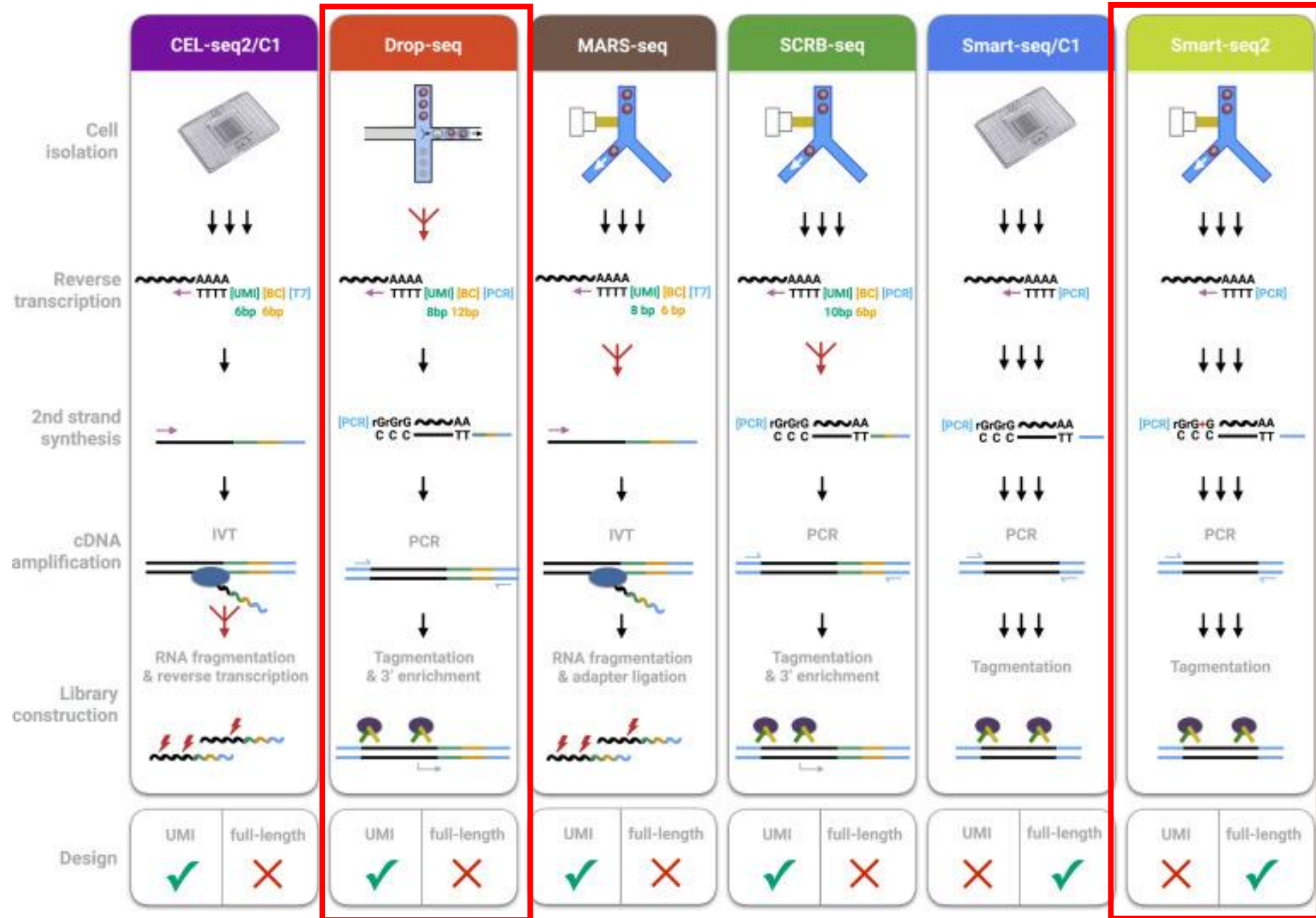


Single Cell

- Why single-cell ?



• scRNA-seq technology

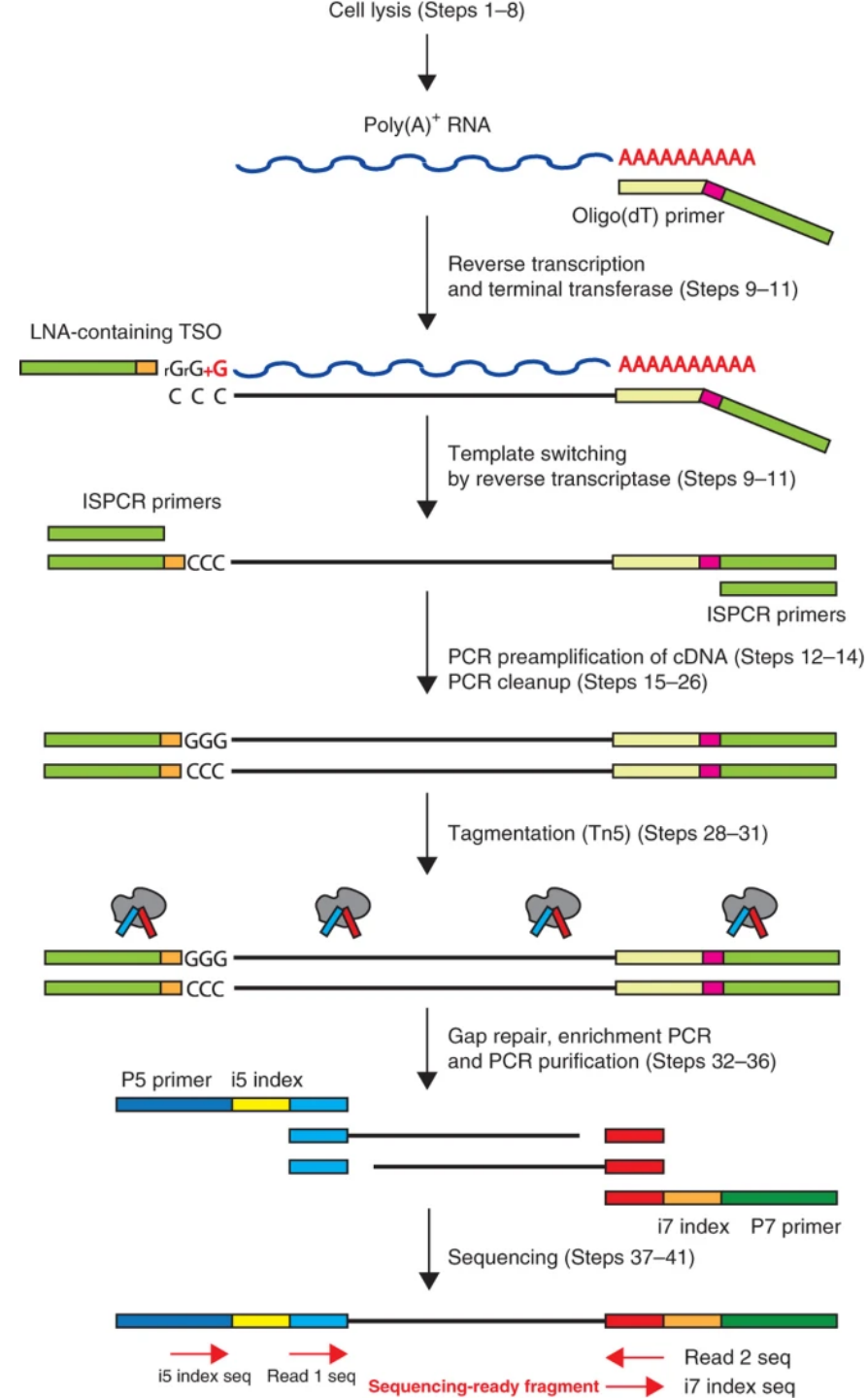


Comparative Analysis of Single-Cell RNA Sequencing Methods

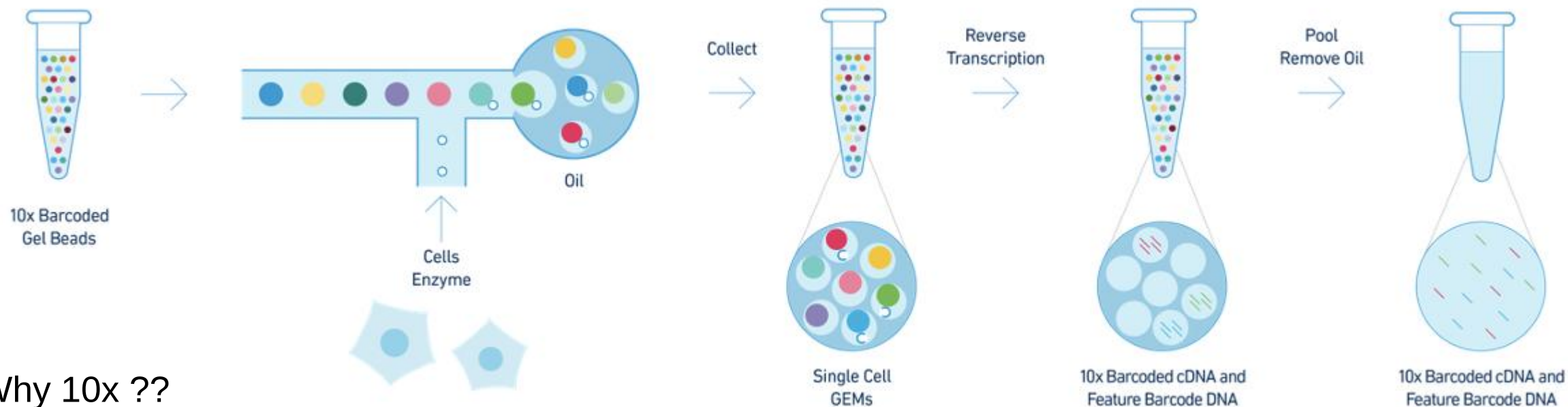
• Smart-seq2

- FACS sorting → 96-well PCR plates
- Cell Lysis
- Reverse transcription → cDNA ...
- Tn5 tagmentation (chopping and tagging)
- Full-length sequencing
- Each Fastq file corresponds to each cell

☐	1	SRR11548680	SAMN14600765	150	185.21 M	66.39 Mb	SRX8118737	GSM4477083	NextSeq 500
☐	2	SRR11548681	SAMN14600764	150	254.55 M	91.26 Mb	SRX8118738	GSM4477084	NextSeq 500
☐	3	SRR11548682	SAMN14600763	150	222.46 M	80.23 Mb	SRX8118739	GSM4477085	NextSeq 500
☐	4	SRR11548683	SAMN14600762	150	224.80 M	81.26 Mb	SRX8118740	GSM4477086	NextSeq 500
☐	5	SRR11548684	SAMN14600761	150	197.79 M	71.15 Mb	SRX8118879	GSM4477087	NextSeq 500
☐	6	SRR11548685	SAMN14600760	150	217.61 M	77.94 Mb	SRX8118880	GSM4477088	NextSeq 500
☐	7	SRR11548686	SAMN14600759	150	285.22 M	101.53 Mb	SRX8118881	GSM4477089	NextSeq 500
☐	8	SRR11548687	SAMN14600758	150	263.85 M	96.43 Mb	SRX8118882	GSM4477090	NextSeq 500
☐	9	SRR11548688	SAMN14600757	150	28.65 k	95.19 kb	SRX8118883	GSM4477091	NextSeq 500
☐	10	SRR11548689	SAMN14600756	150	274.38 M	98.97 Mb	SRX8118884	GSM4477092	NextSeq 500
☐	11	SRR11548690	SAMN14600755	150	214.02 M	77.81 Mb	SRX8118885	GSM4477093	NextSeq 500
☐	12	SRR11548691	SAMN14600754	150	258.18 M	93.43 Mb	SRX8118886	GSM4477094	NextSeq 500
☐	13	SRR11548692	SAMN14600753	150	193.40 M	69.72 Mb	SRX8118887	GSM4477095	NextSeq 500
☐	14	SRR11548693	SAMN14600752	150	104.79 M	37.42 Mb	SRX8118888	GSM4477096	NextSeq 500
☐	15	SRR11548694	SAMN14600751	150	152.84 M	55.17 Mb	SRX8118889	GSM4477097	NextSeq 500



- 10x technology (drop-seq based)



Why 10x ??

→ High throughput (for cell point of view)
& cheaper per cell

-Drawback: since droplet size is fixed and the cell must go inside the droplet, there is a physical restriction for a cell to enter the droplet (Big cell cannot go inside)

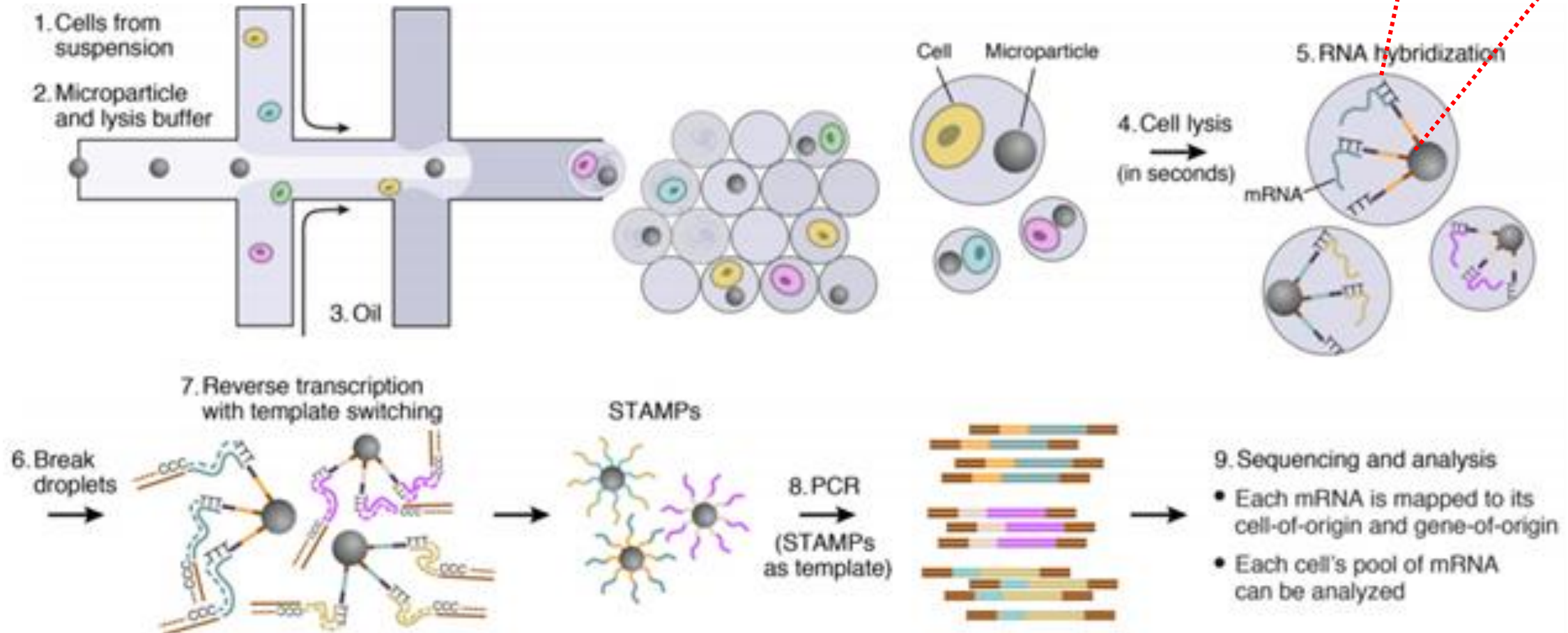
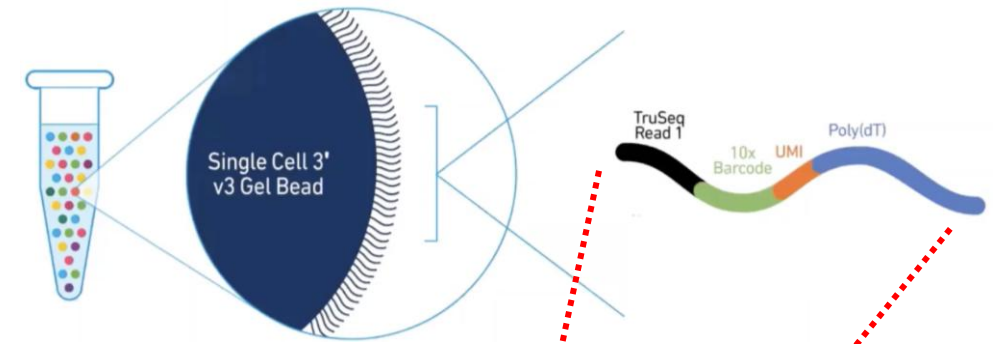
-Cell dissociation stress: hard to observe epithelial cells or stroma cells
Bias in liquid cell: Immune cells

☒	☐	Run	BioSample	Bases	Bytes	egfr_status
☐	1	SRR11040644	SAMN14057011	21.96 G	14.67 Gb	wild type
☐	2	SRR11040645	SAMN14057010	20.42 G	14.40 Gb	mutation
☐	3	SRR11040646	SAMN14057009	21.22 G	13.77 Gb	mutation
☐	4	SRR11040647	SAMN14057008	20.82 G	13.43 Gb	wild type
☐	5	SRR11040648	SAMN14057007	24.77 G	16.94 Gb	mutation
☐	6	SRR11040649	SAMN14057006	23.37 G	16.38 Gb	wild type
☐	7	SRR11040650	SAMN14057005	24.45 G	17.20 Gb	mutation
☐	8	SRR11040651	SAMN14057004	26.17 G	18.69 Gb	mutation
☐	9	SRR11040652	SAMN14057003	22.82 G	15.15 Gb	wild type
☐	10	SRR11040653	SAMN14057002	22.43 G	14.91 Gb	wild type

1 fastq file (result of single experiment)
→ ~3k ~ 10K cells!

• 10x technology

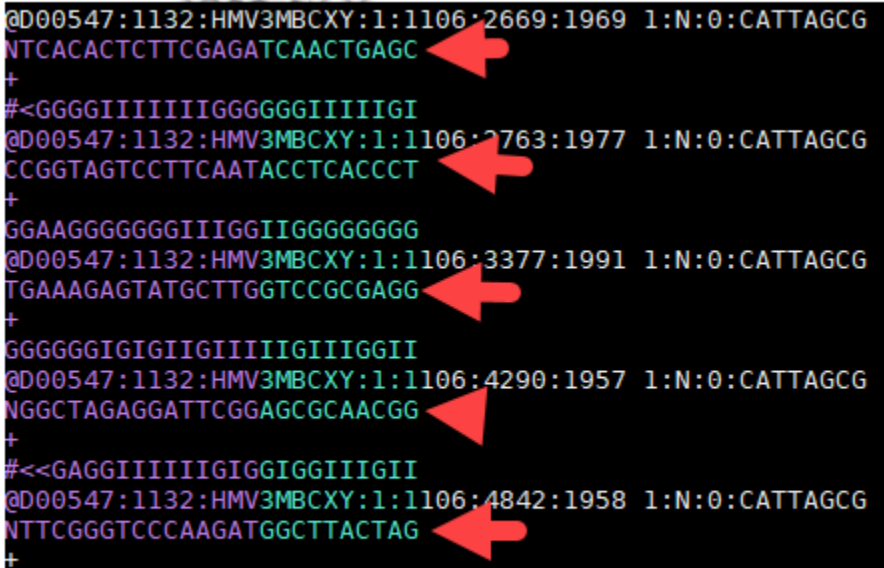
- 10x barcode → distinguish each cell
- UMI: avoid PCR amplification bias → transcript counting
- Poly(dT): captures poly A tail from mRNA



• Raw data

-Of course, FASTQ file

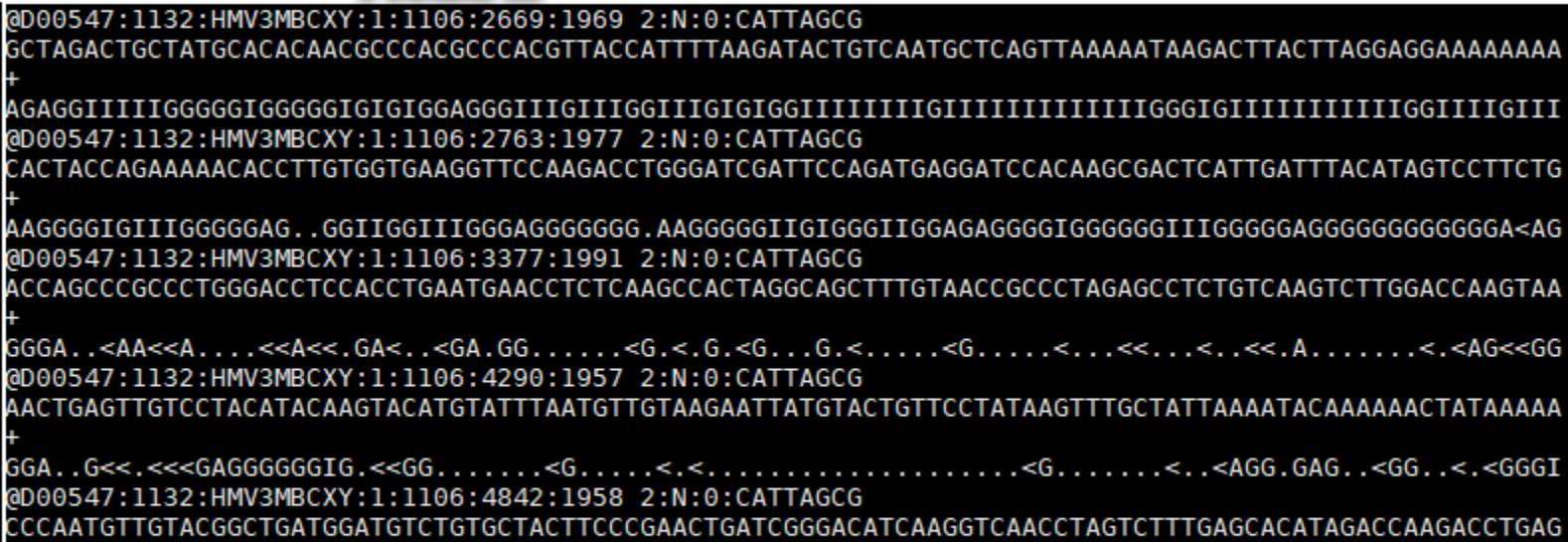
Read 1



Cell
Barcode

UMI

Read 2



mRNA

-Read1 (or I file): Cell barcode & UMI sequence

Variety of UMI → gene expression

-Read2: mRNA sequencing information → STAR → which gene?

But limited length; biased to 3' region

Technically paired-end sequencing → Biologically single-end sequencing

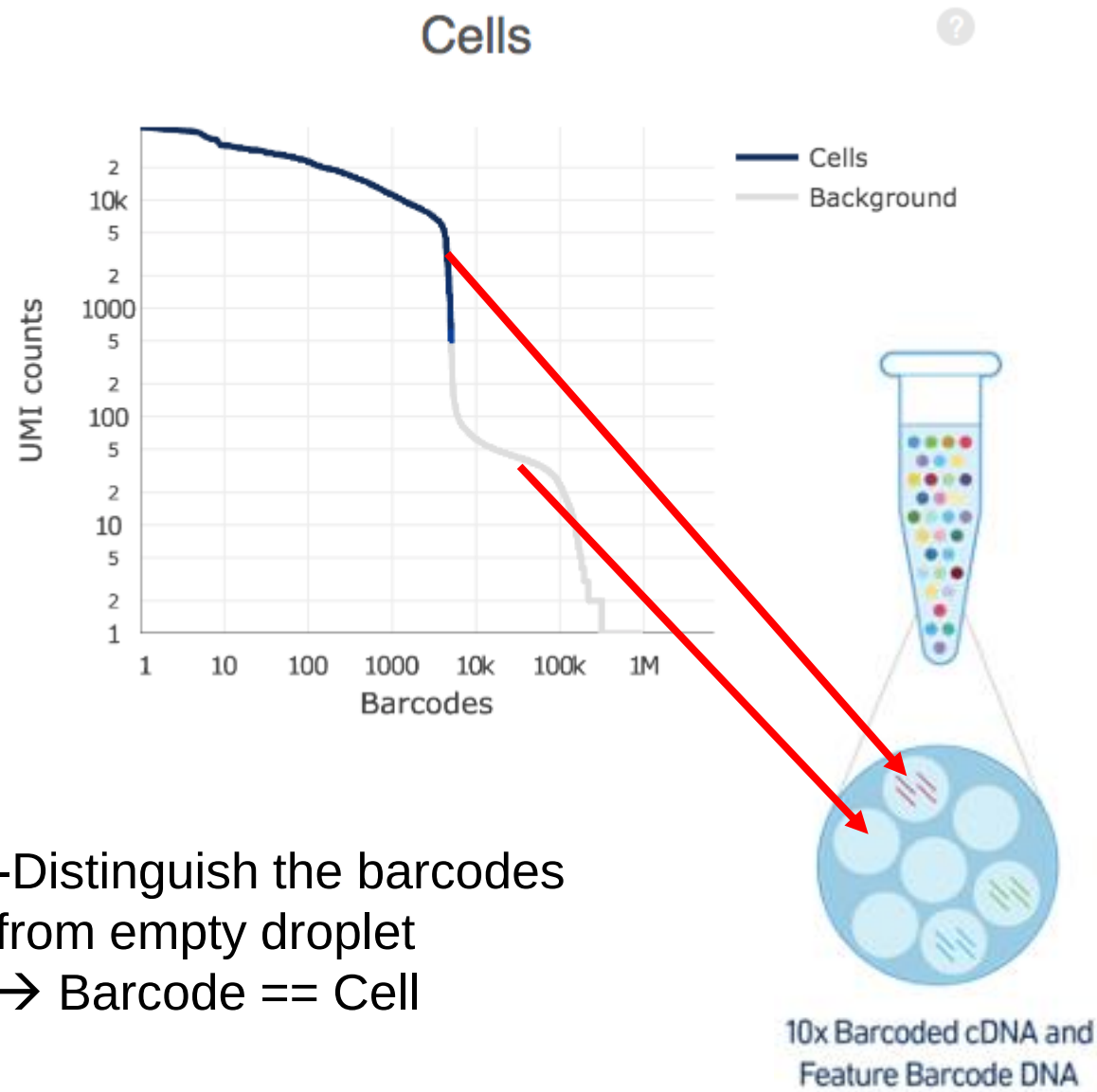
- Raw data

-Single cell → Transcript is scarce → low capture rate → high drop out (many zero count: 90 ~ 95 %)
+ short read (tend to be multi-mapped), single-end (low confidence)
!! Hard to distinguish between real zero and drop out

	Cell1	Cell2	...	CellN
<i>Gene1</i>	3	2	.	13
<i>Gene2</i>	2	3	.	1
<i>Gene3</i>	1	14	.	18
...	.	.	.	.
...	.	.	.	.
...	.	.	.	.
<i>GeneM</i>	25	0	.	0

	Cell1	Cell2	...	CellN
<i>Gene1</i>	3			
<i>Gene2</i>		3	.	
<i>Gene3</i>		14		18
...	.		.	.
...				
...	.	.	.	
<i>GeneM</i>	25	0		0

Initial quality control



-Distinguish the barcodes from empty droplet
→ Barcode == Cell

Summary

Gene Expression

Antibody

125

Estimated Number of Cells

3,200

Mean Reads per Cell

13

Median Genes per Cell

Sequencing

Number of Reads	400,000
Number of Short Reads Skipped	0
Valid Barcodes	94.0%
Valid UMIs	99.9%
Sequencing Saturation	75.0%
Q30 Bases in Barcode	96.4%
Q30 Bases in RNA Read	95.7%
Q30 Bases in UMI	96.3%

Mapping

Reads Mapped to Genome	100.0%
Reads Mapped Confidently to Genome	21.4%
Reads Mapped Confidently to Intergenic Regions	2.6%
Reads Mapped Confidently to Intronic Regions	12.5%
Reads Mapped Confidently to Exonic Regions	6.3%
Reads Mapped Confidently to Transcriptome	16.3%
Reads Mapped Antisense to Gene	2.0%

Cells

Failed

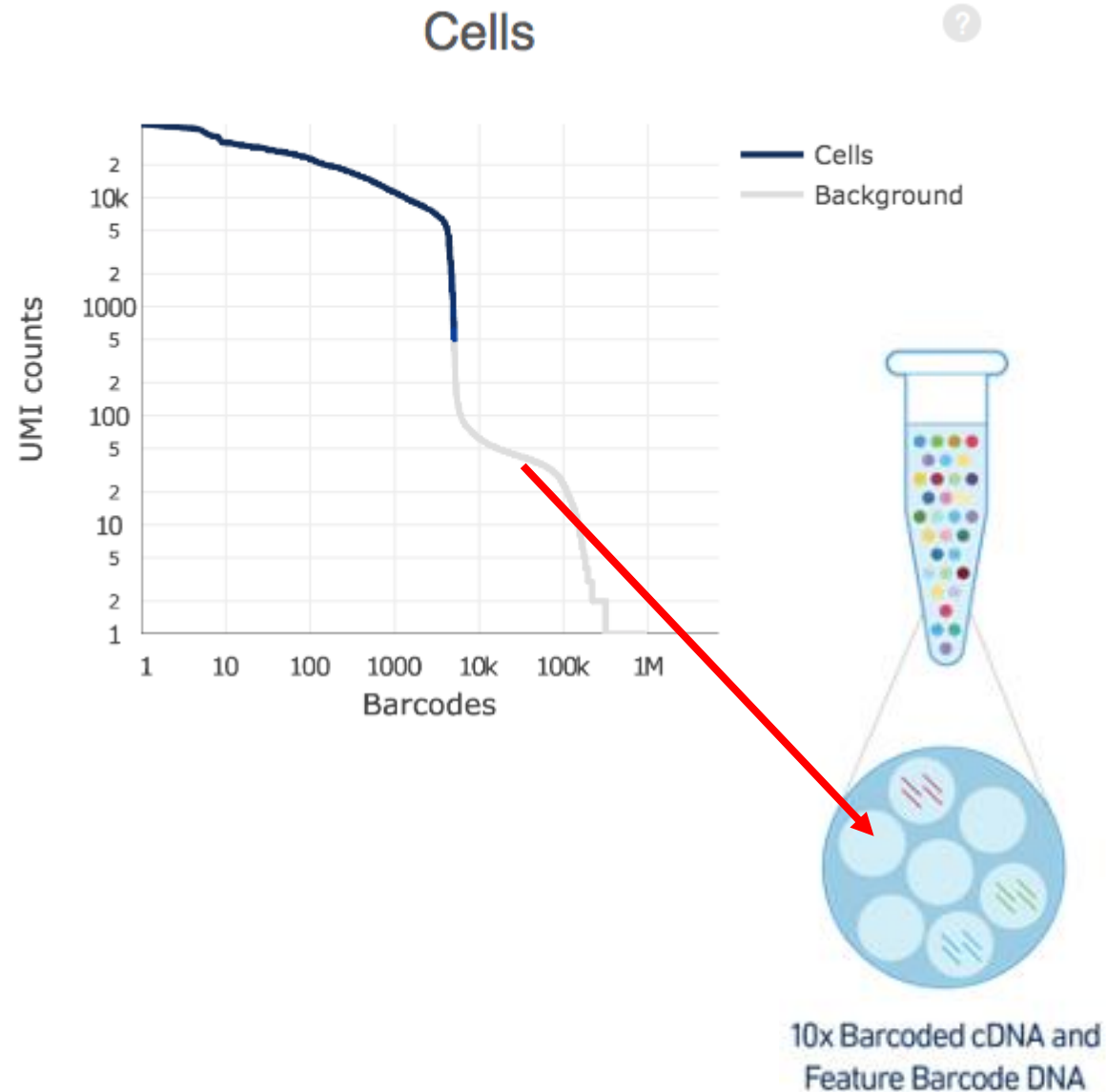
Barcode Rank Plot

Estimated Number of Cells	125
Fraction Reads in Cells	49.1%
Mean Reads per Cell	3,200
Median UMI Counts per Cell	46
Median Genes per Cell	13
Total Genes Detected	78

Sample

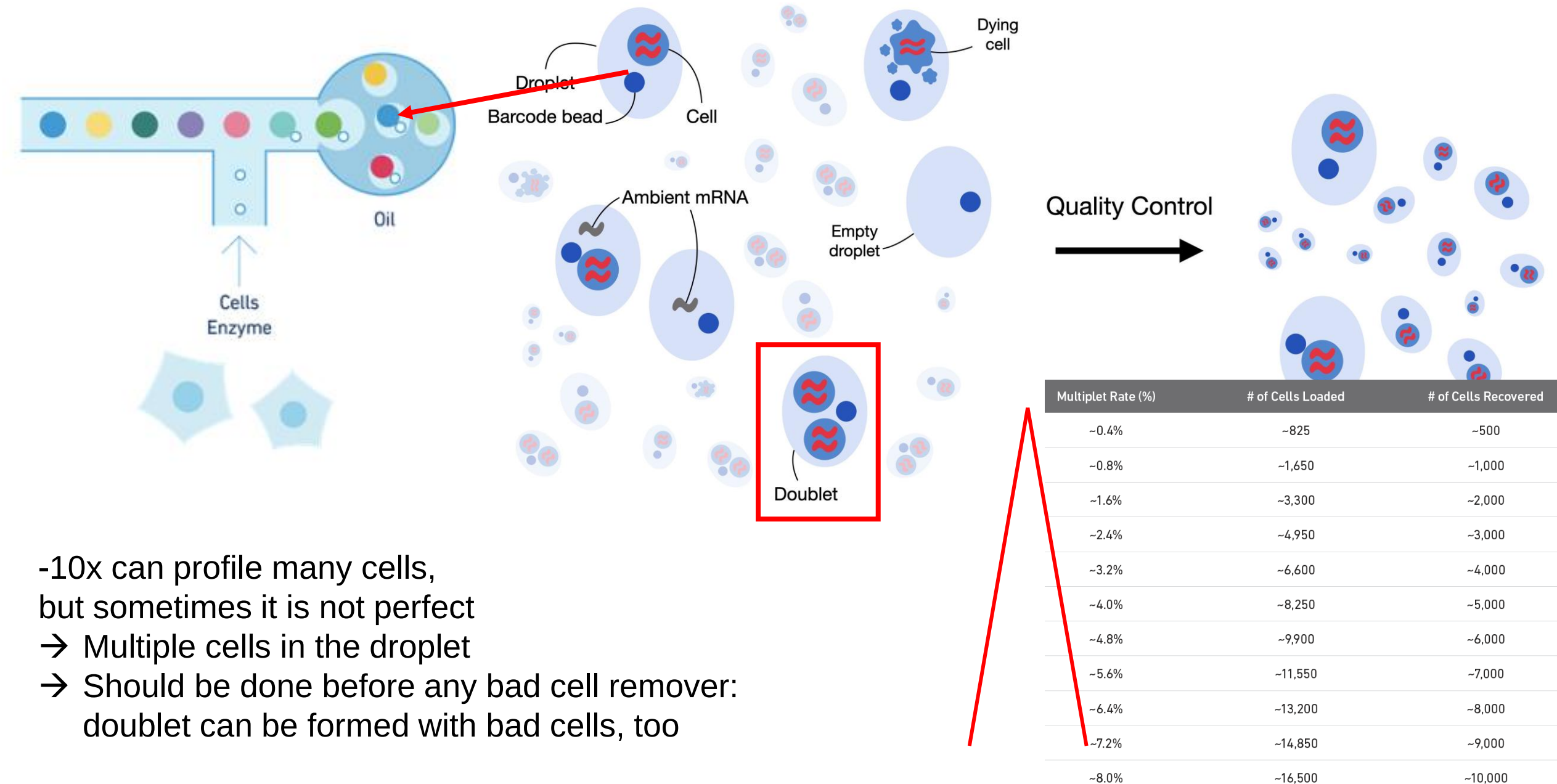
Sample ID	78388_chr21_400K
Sample Description	
Chemistry	Single Cell 3' v3
Include introns	True
Reference Path	...refdata-cellranger-chr21-3.0.0
Transcriptome	GRCh38_chr21-3.0.0
Pipeline Version	7.0.0

• Ambient RNA



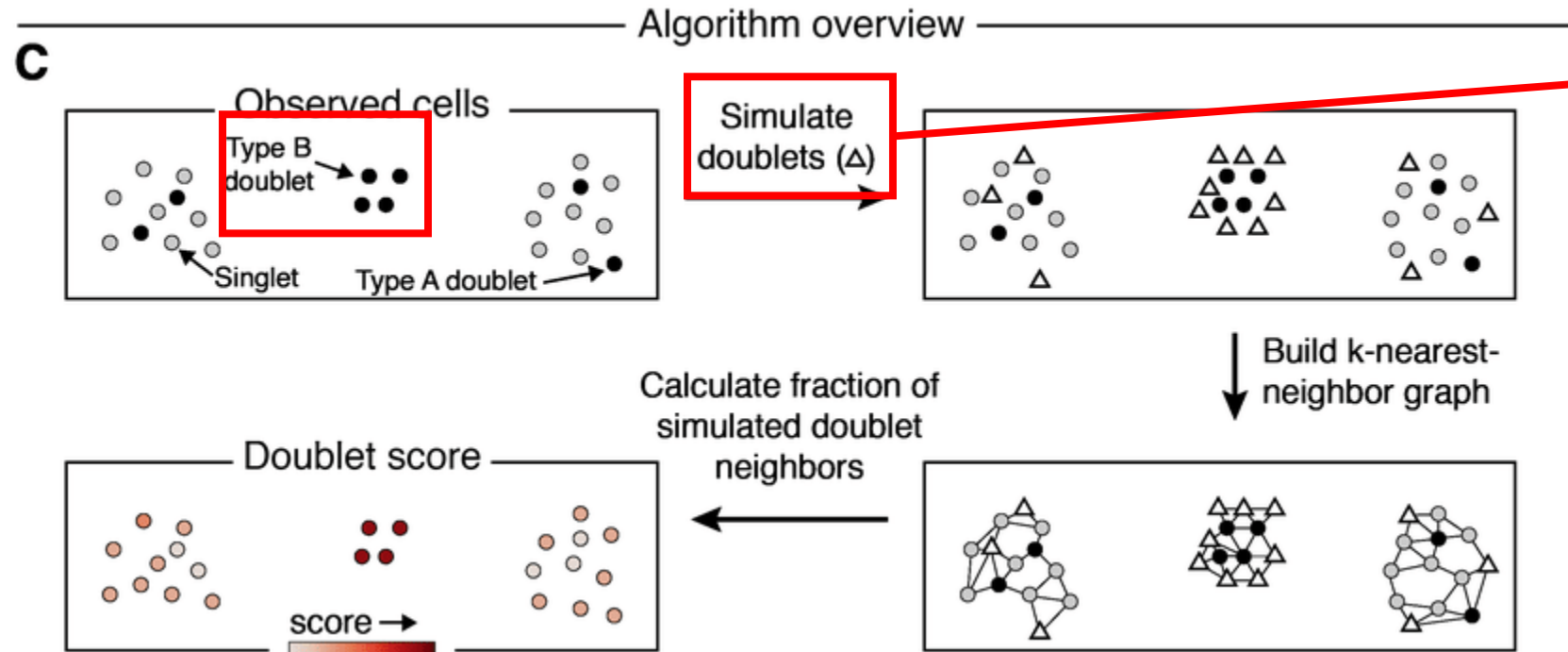
- There is still some RNA detection from the empty droplet
- This ambient RNA might be universal for all droplets, even those that contain a cell
- Cellbender (at the sequence level), SoupX (at the count level)
- Training data set: empty droplet
 - Adjust the ambient RNA distribution to real cells
 - Adjust the expression values for each cell

• Doublet (multiplet) detection



- 10x can profile many cells, but sometimes it is not perfect
- Multiple cells in the droplet
- Should be done before any bad cell remover: doublet can be formed with bad cells, too

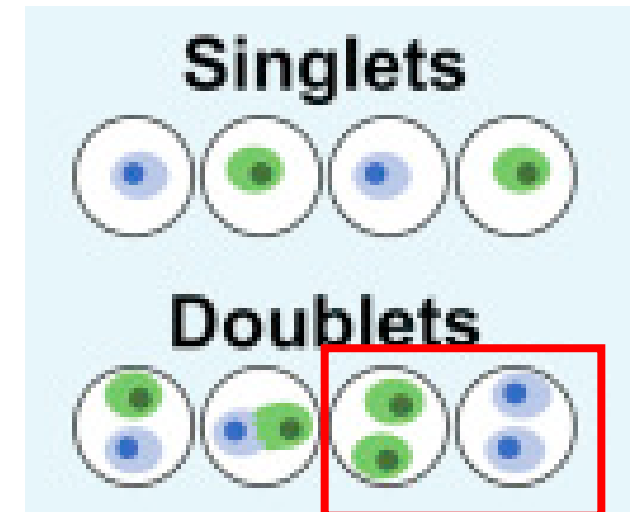
- Doublet (multiplet) detection



-make a synthetic doublets by merging two cells

-compare those synthetic doublets with Singlets and Doublets

DoubletFinder, Scrublet, scDbIFinder, ...



But it is still hard to distinguish the homotropic doublets

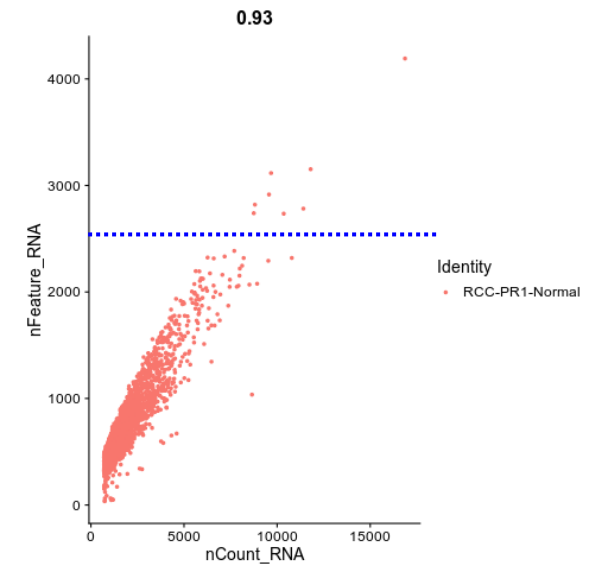
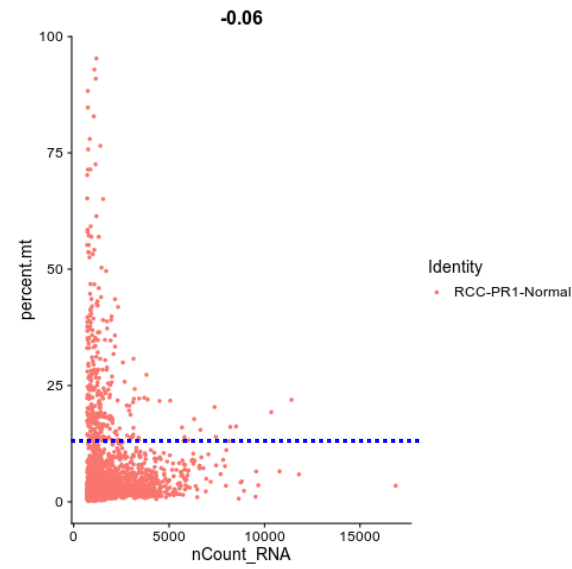
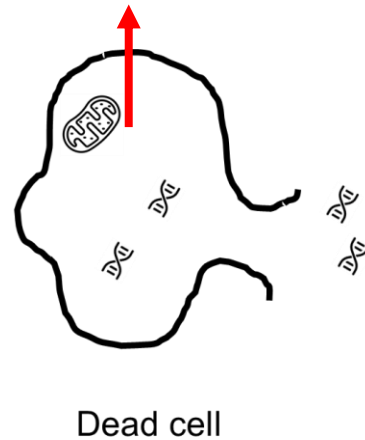
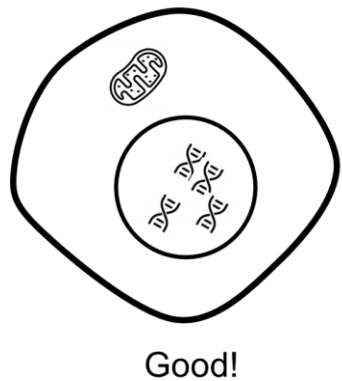
• Quality control for bad cells

-Quality control

Why? There might be some bad cells collected
(reason: cell stress during processing, high drop-out)

Low nCount or nFeature → empty droplet
nFeature > 150 ~ 200 (There should be certain number of gene detected: Housekeeping gene)

High nCount or nFeature → doublet
High MT % → dead cell



- Quality control for bad cells

-Quality control

Oops!

1) Neutrophils (or other granulocytes): relatively low RNA content and relatively high levels of RNases and other inhibitory compounds, resulting in fewer transcripts detected
→ They need to secrete cytolytic enzymes, no time for other gene expression

2) Plasma cell: relatively low RNA content
→ They need to secrete antibodies!

3) Red blood cell: low RNA content and nFeatures
→ It has no nucleus; no transcription (But we usually don't analyze it)

• Normalization and Scaling

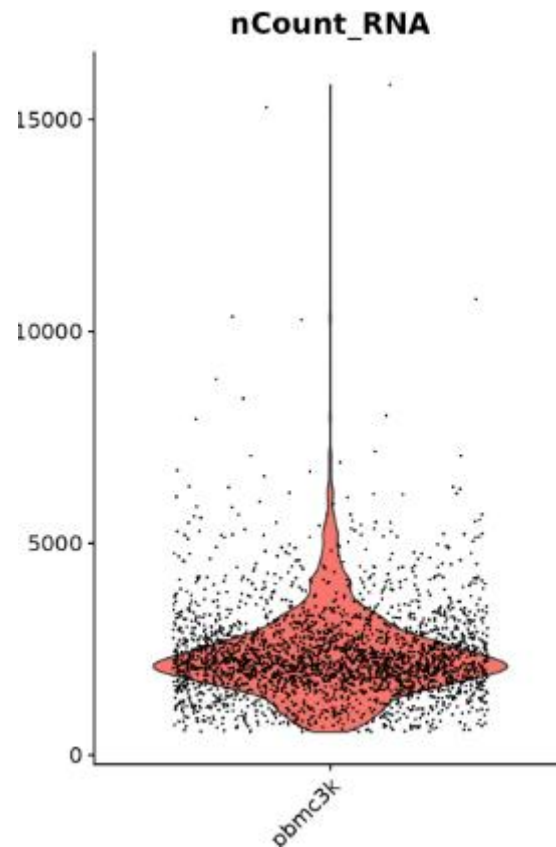
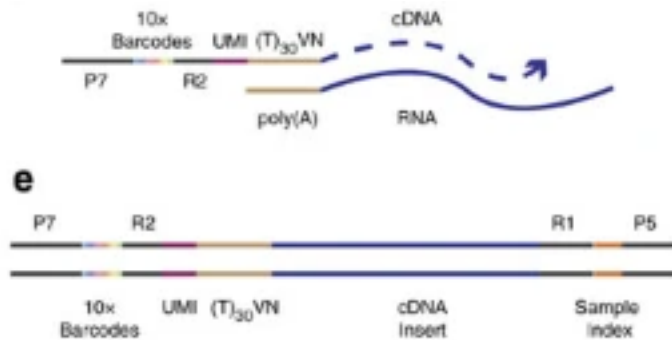
-Total read count normalization: Adjust read-depth between each cell

Log-transformation: adjust the variance of the gene expression matrix

Scaling: gives equal weight in downstream analyses, so that highly-expressed genes do not dominate

!! No gene length normalization

→ Only captures a short region of 3' end → no bias for gene length since every gene has only one 3' end



Broad range of read depth for each cell

→ Different distribution of gene expression for each cell

- **sctransform normalization**

- Each cell is heterogeneous

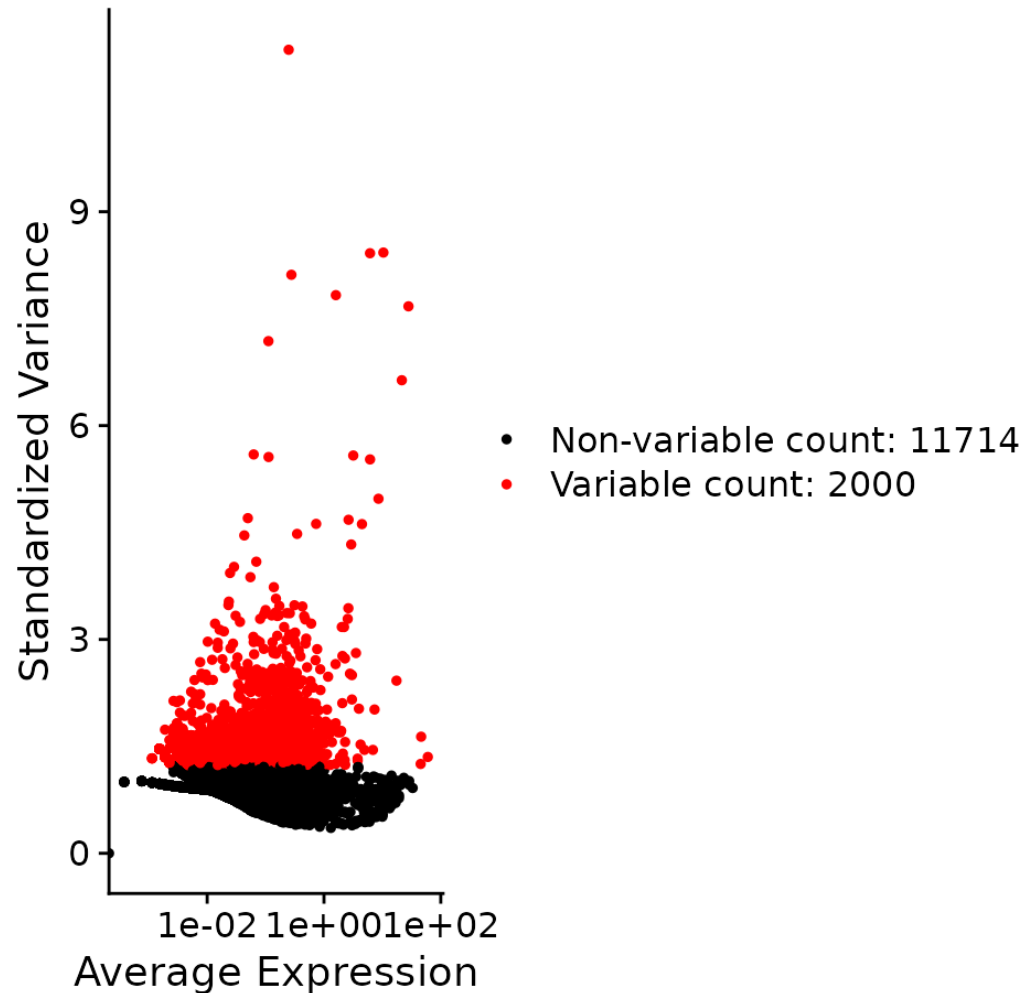
- Confounded by technical factors (sequencing depth)

- Cell cycling phase

- sctransform regresses out those confounding effects (using generalized linear model)

• Feature selection

- scRNA-seq tends to have a lot of drop-out
 - Not all the genes are informative → select informative genes and reduce noise
 - Highly variable genes (across different cells)



- In bioinformatics (or data analysis point of view), Variance can reflect the amount of information
- It is likely that highly expressed genes (high mean value) have high variance
 - therefore, we must adjust variance by mean value
- Common red flag hkg genes
 - HLA, TCR/BCR (individual diversity), cell cycling (we don't want cell cycling phase affect the cell annotation)
- !! We always need to consider individual (batch) specific genes (to remove)

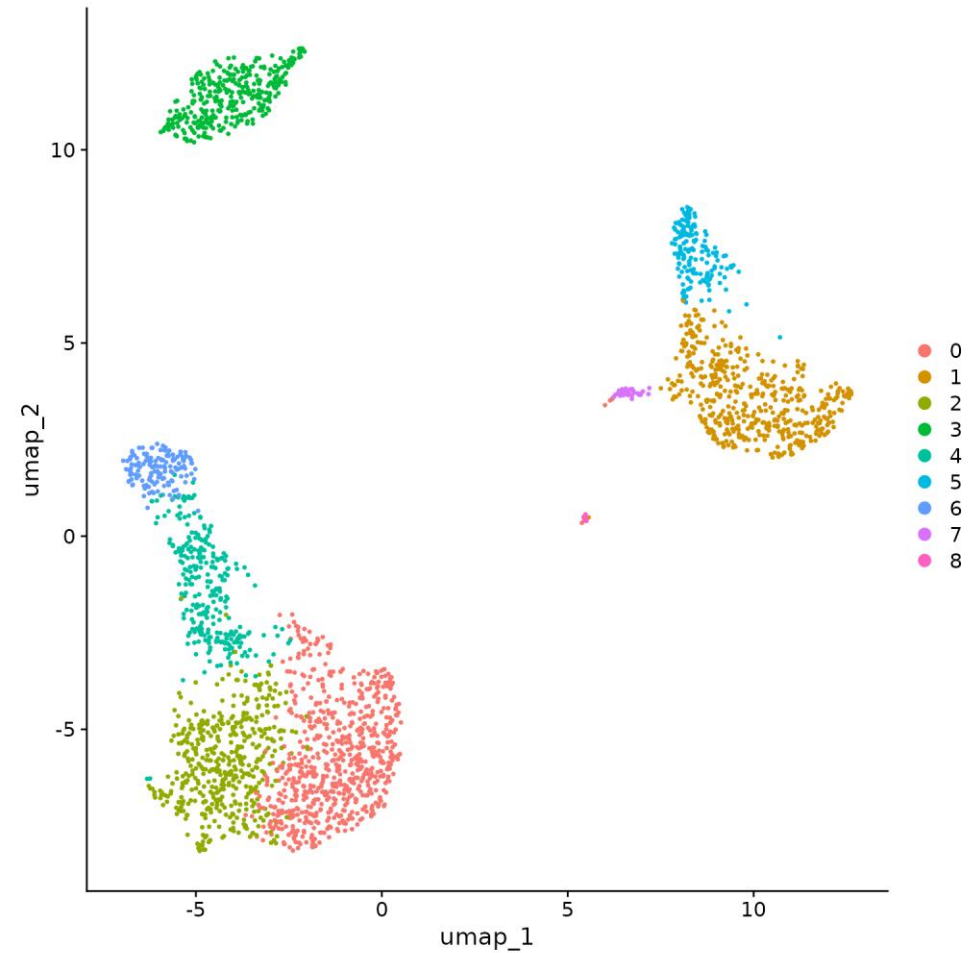
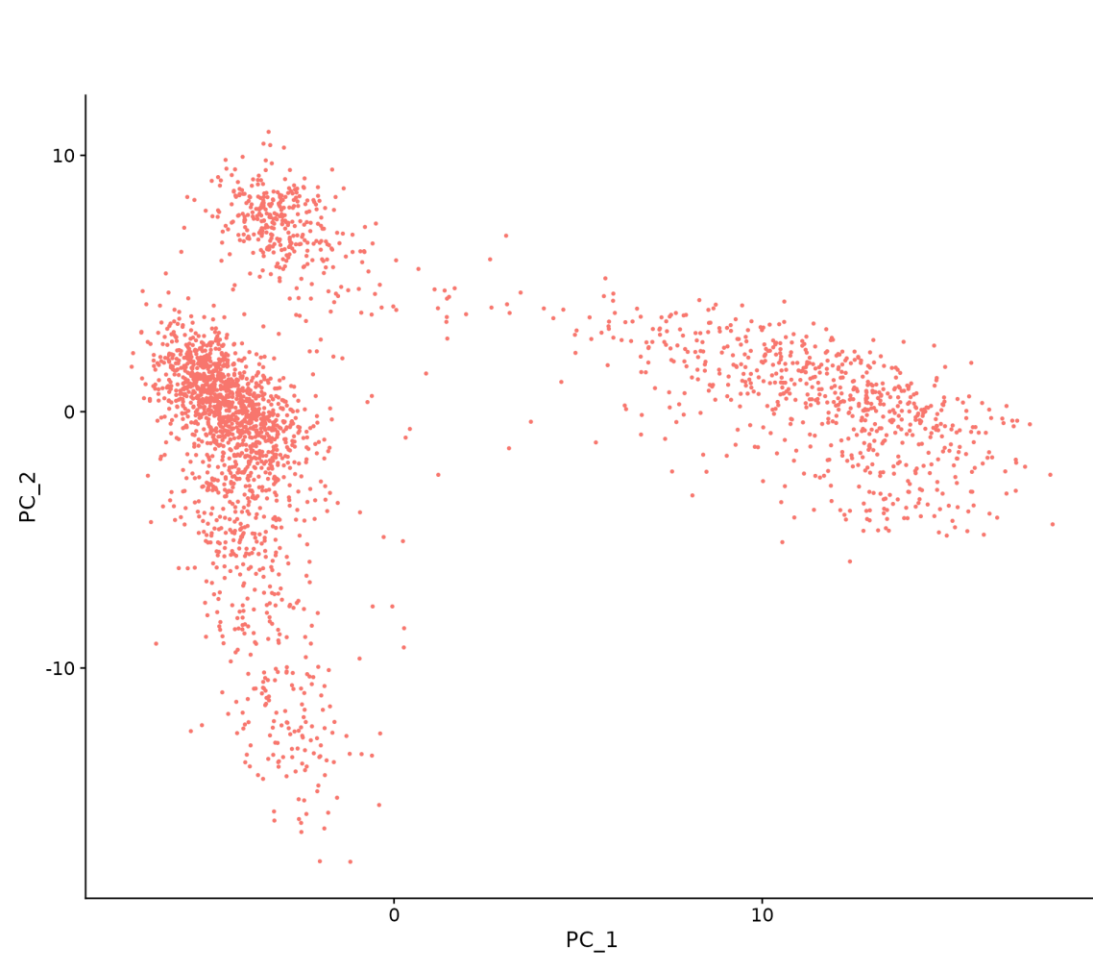
• Dimension reduction (PCA & UMAP)

-Too many features (genes) → hard to interpret

→ Dimension reduction: abstract of many features!

PCA: commonly used in bulk RNA-seq data → insufficient for scRNA-seq

UMAP: adjusted for scRNA-seq (similar cells to be close and different cells to be far away)



• Batch correction (Harmony)

-Reason → To remove technical variation or confounding effect between samples

-Assessment

Batch: good mixing

Cell type (or cluster): separated

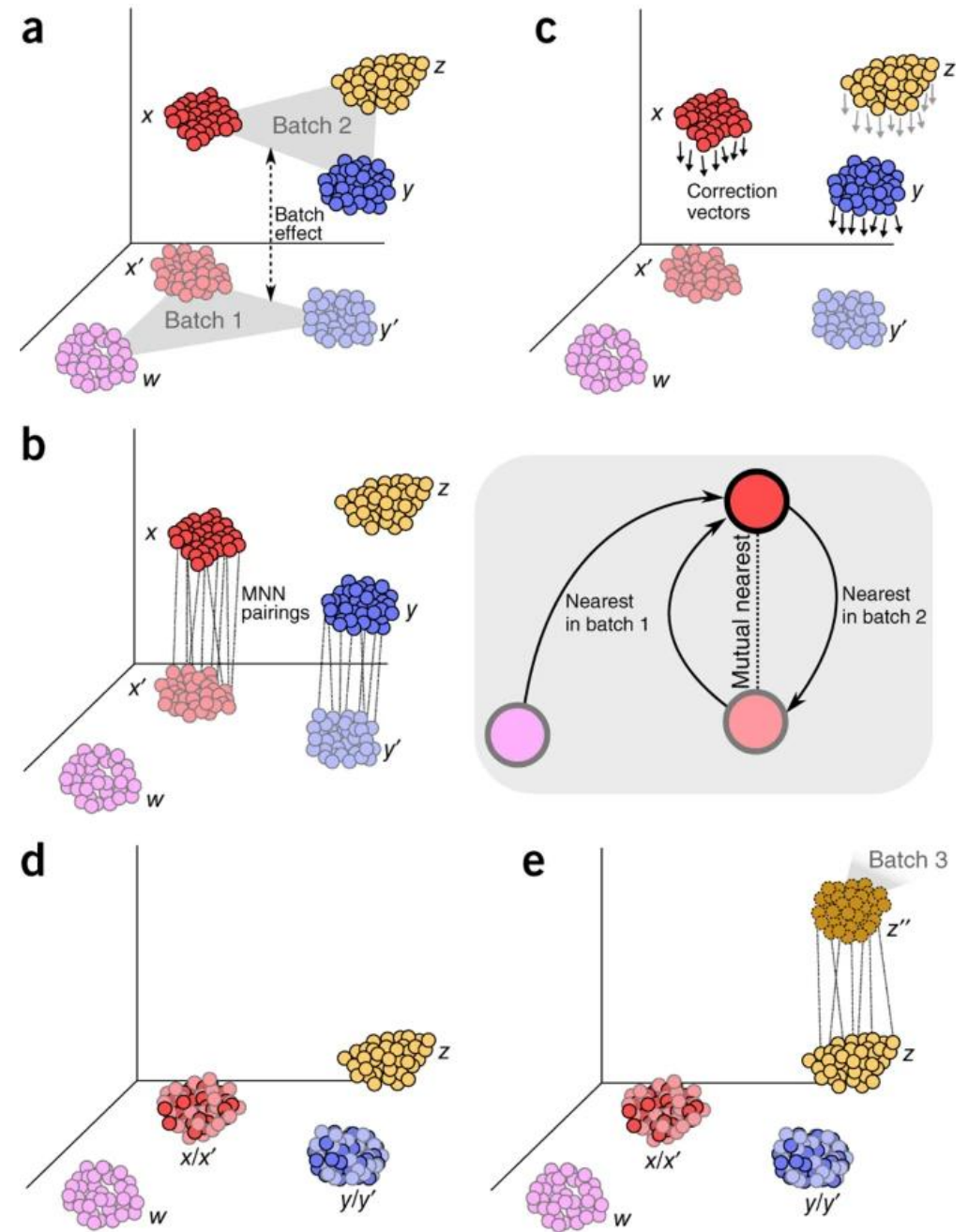
-Entropy: $S = - \sum_{i=1}^n p_i \ln p_i = \ln n$ (High: mixing)

-Silhouette Coefficient: $s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$

b: distance to closest neighbors

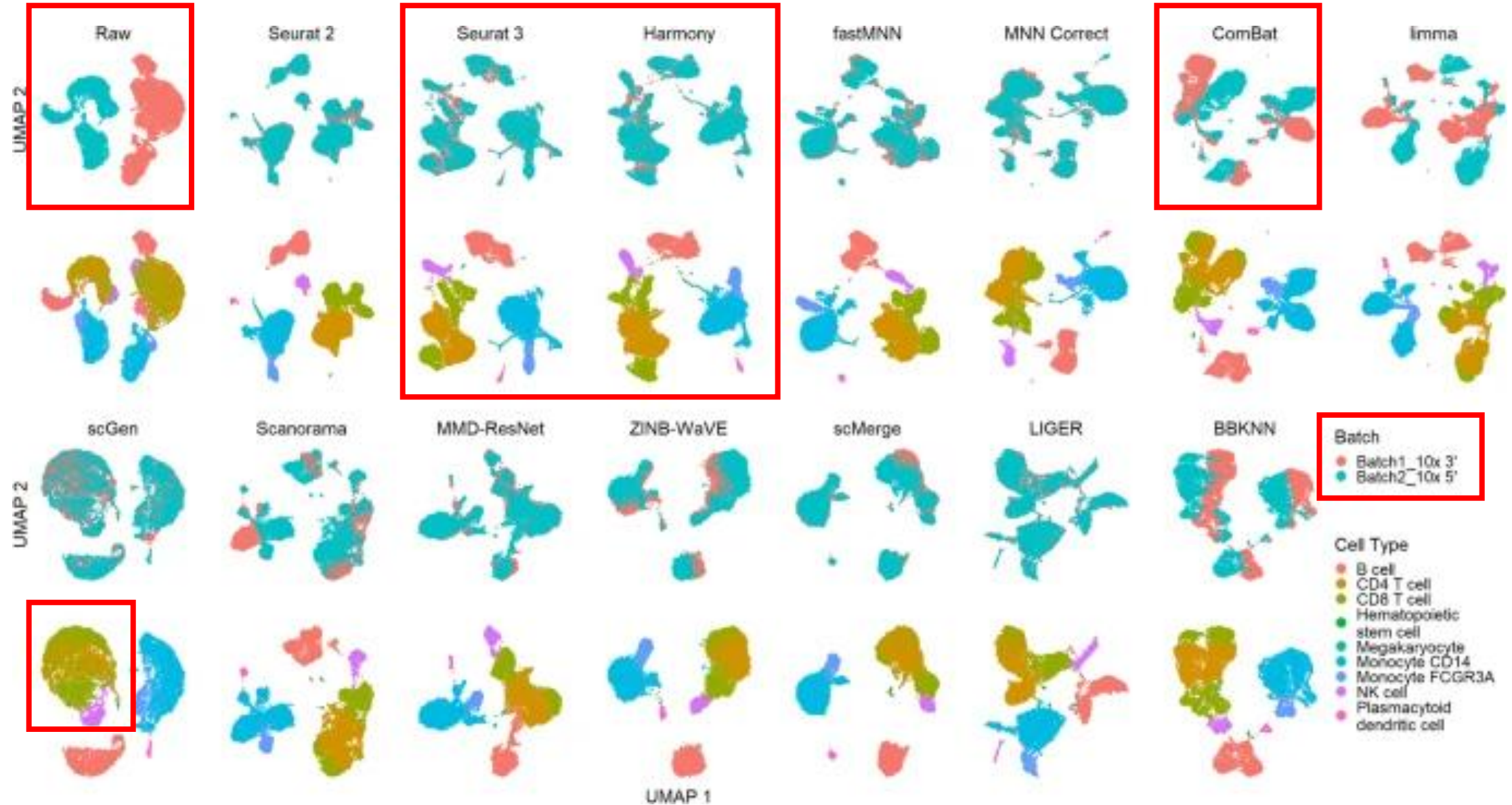
a: distance to self cluster

s → high: far from neighbors → separated



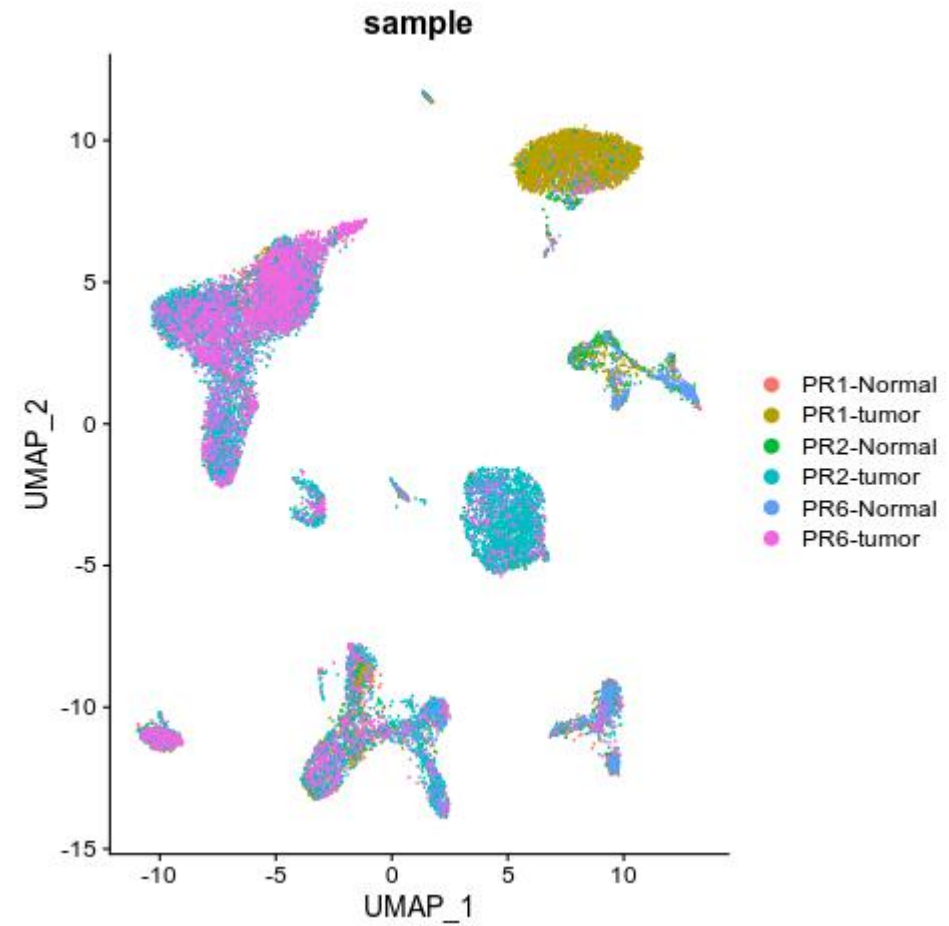
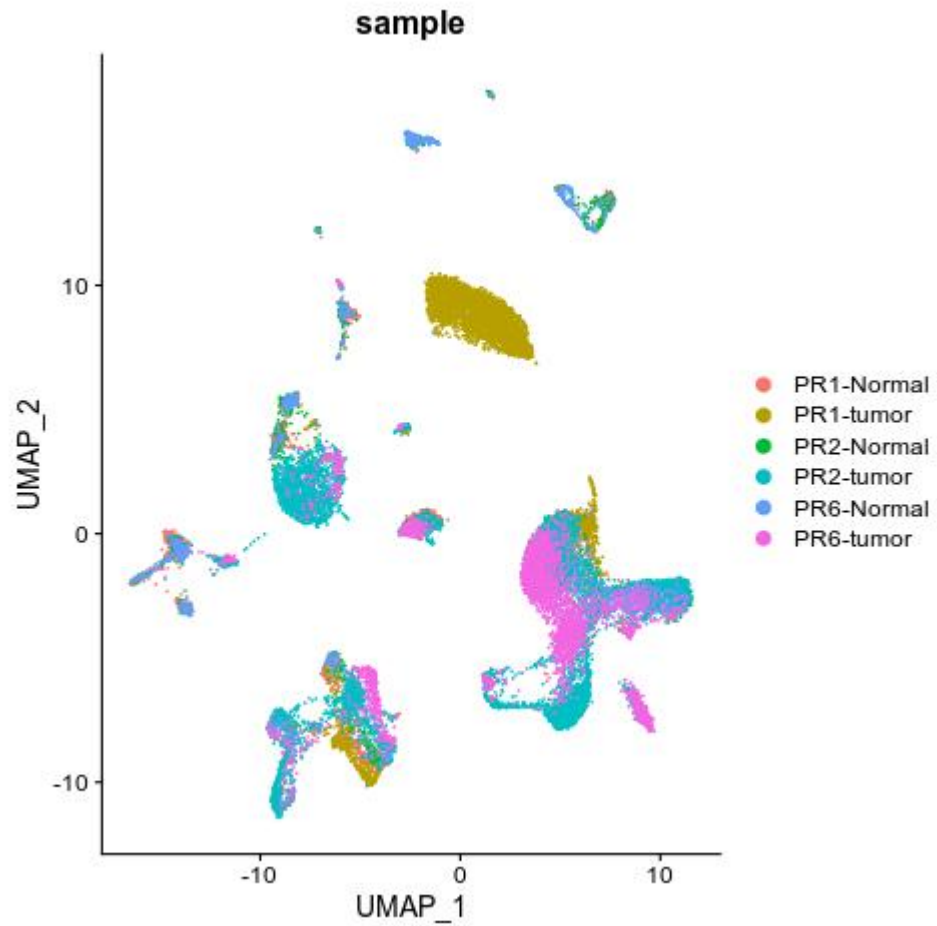
Batch effects in single-cell RNA-sequencing data are corrected by matching mutual nearest neighbors

- Batch correction (Harmony)



A benchmark of batch-effect correction methods for single-cell RNA sequencing data

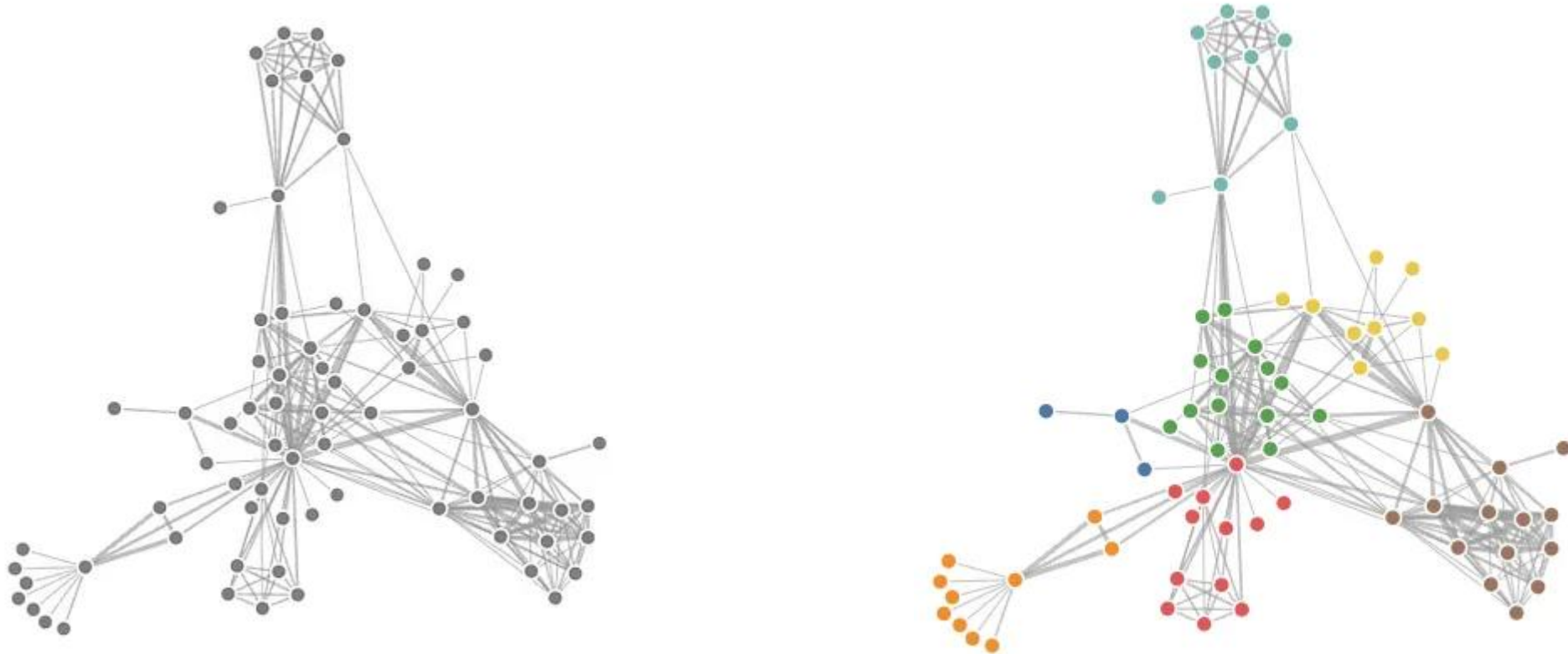
- Batch correction (Harmony)
- Sample level batch correction



- **Clustering**

- Louvain clustering: considering the modularity of the (cell) graph

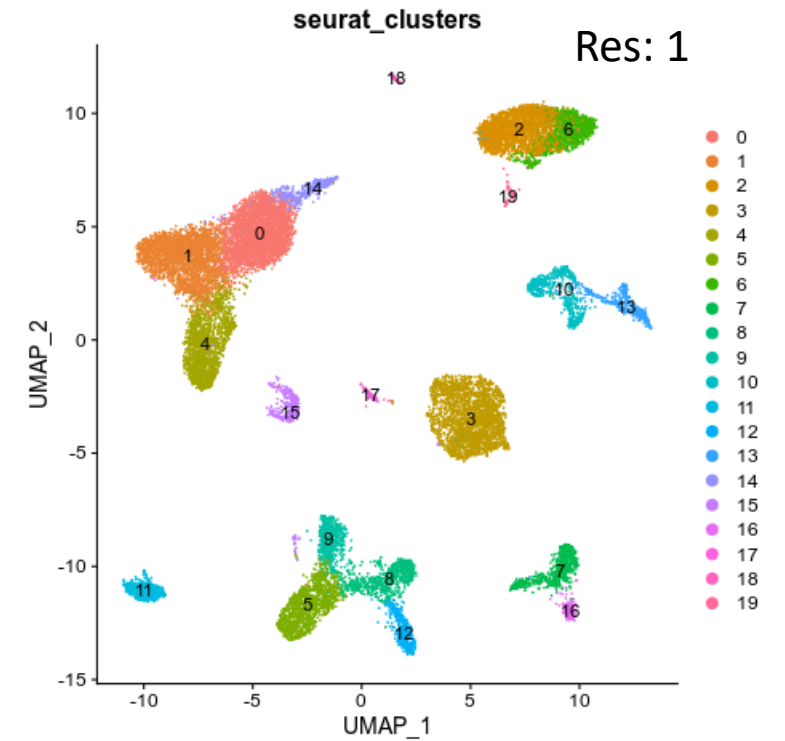
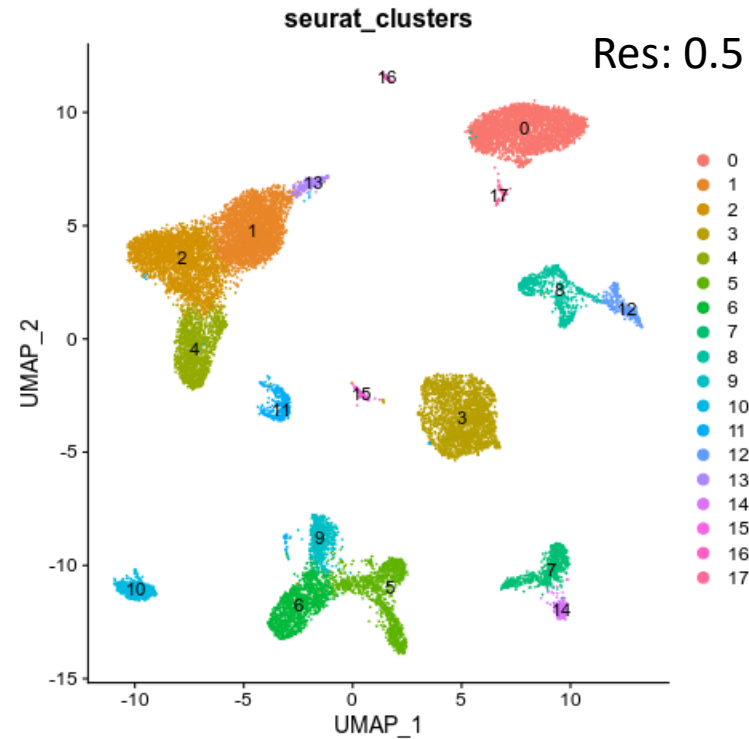
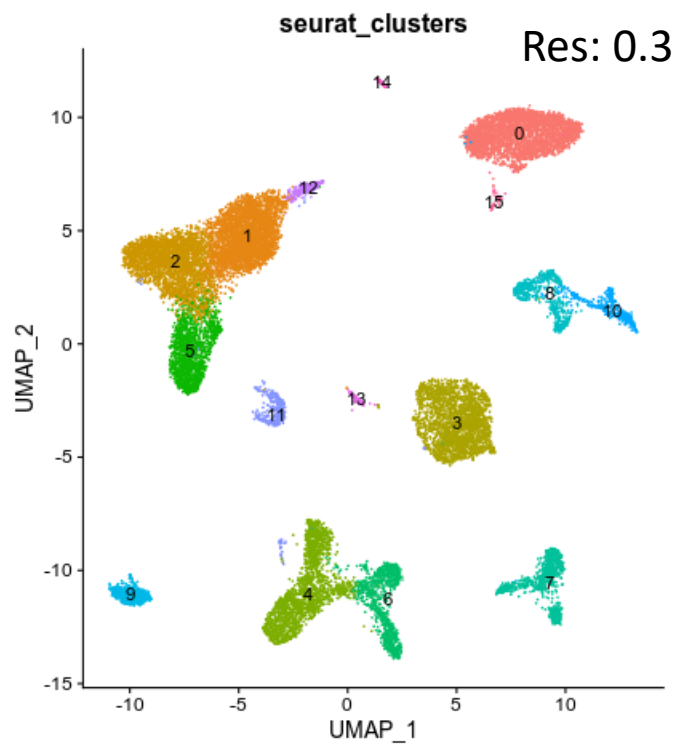
Before performing the clustering, Seurat package obtained SNN (shared-nearest neighbor) graph



- Clustering

- Louvain clustering

Resolution → controls the number of clusters



- Celltype annotation

How? Distribution of marker gene expression

-Functional marker: CD3 for T cells

-Expression marker: MHC class2 for T cells

* Immune cell

Tcell: "CD3D", "CD3E"

CD4 T cell: "CD4"

CD8 T cell: "CD8A", "CD8B"

Treg: "FOXP3", "IL2RA"

NK cell: "KLRB1", "GNLY", "KLRD1", "NKG7"

B cell: "MS4A1", "CD79B"

Macrophage: "C1QA", "C1QB", "CD14", "CD68"

Monocyte: "FCN1", "S100A8", "S100A9"

Mast: "TPSAB1", "CPA3"

Cycling: "MKI67", "TOP2A"

* Non-immune cell

Pericyte: "CSPG4", "MCAM", "MYH11"

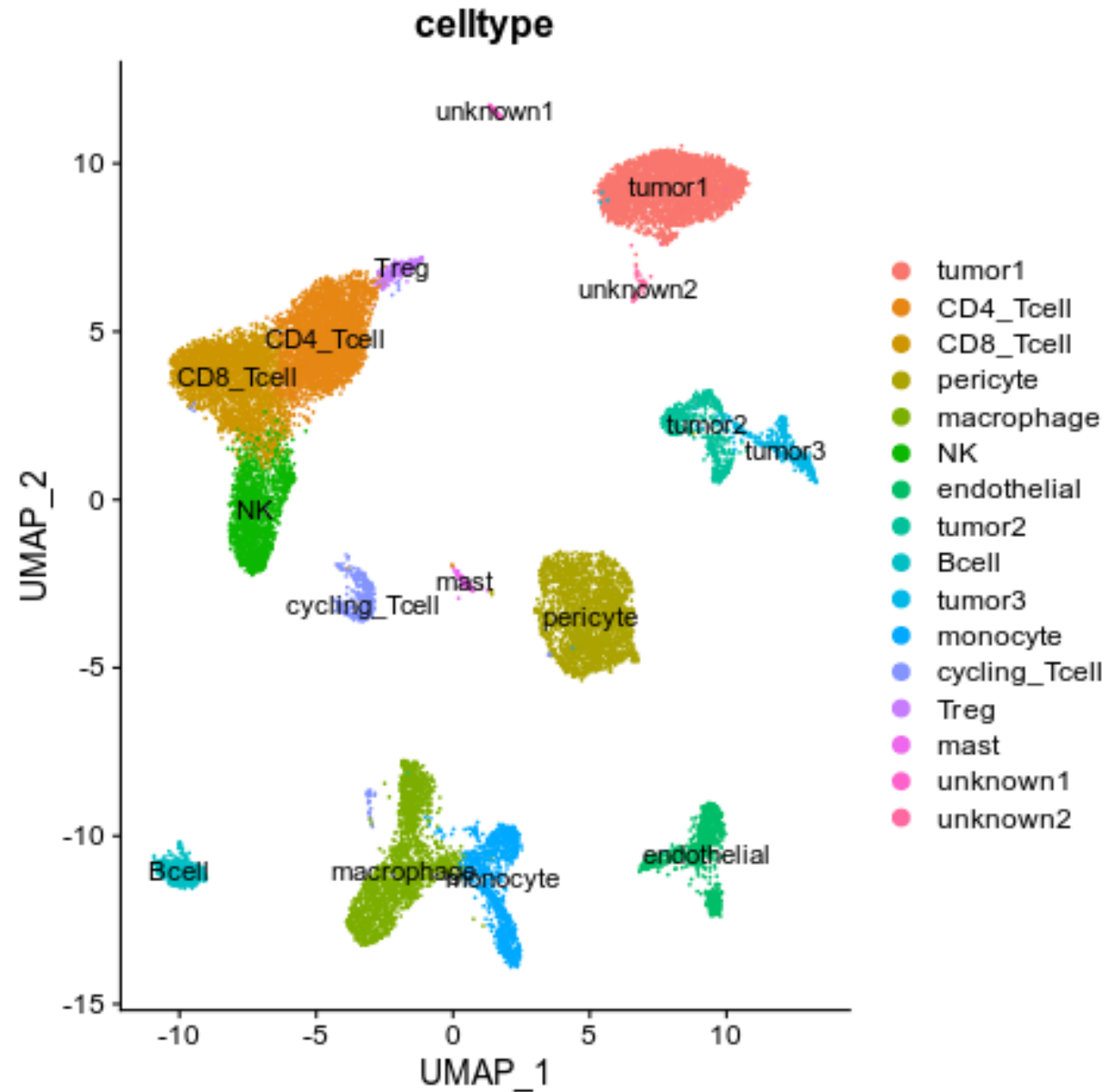
Endothelial cell: "RAMP2", "RNASE1", "ENG", "EGFL7"

Cancer: "PAX8"

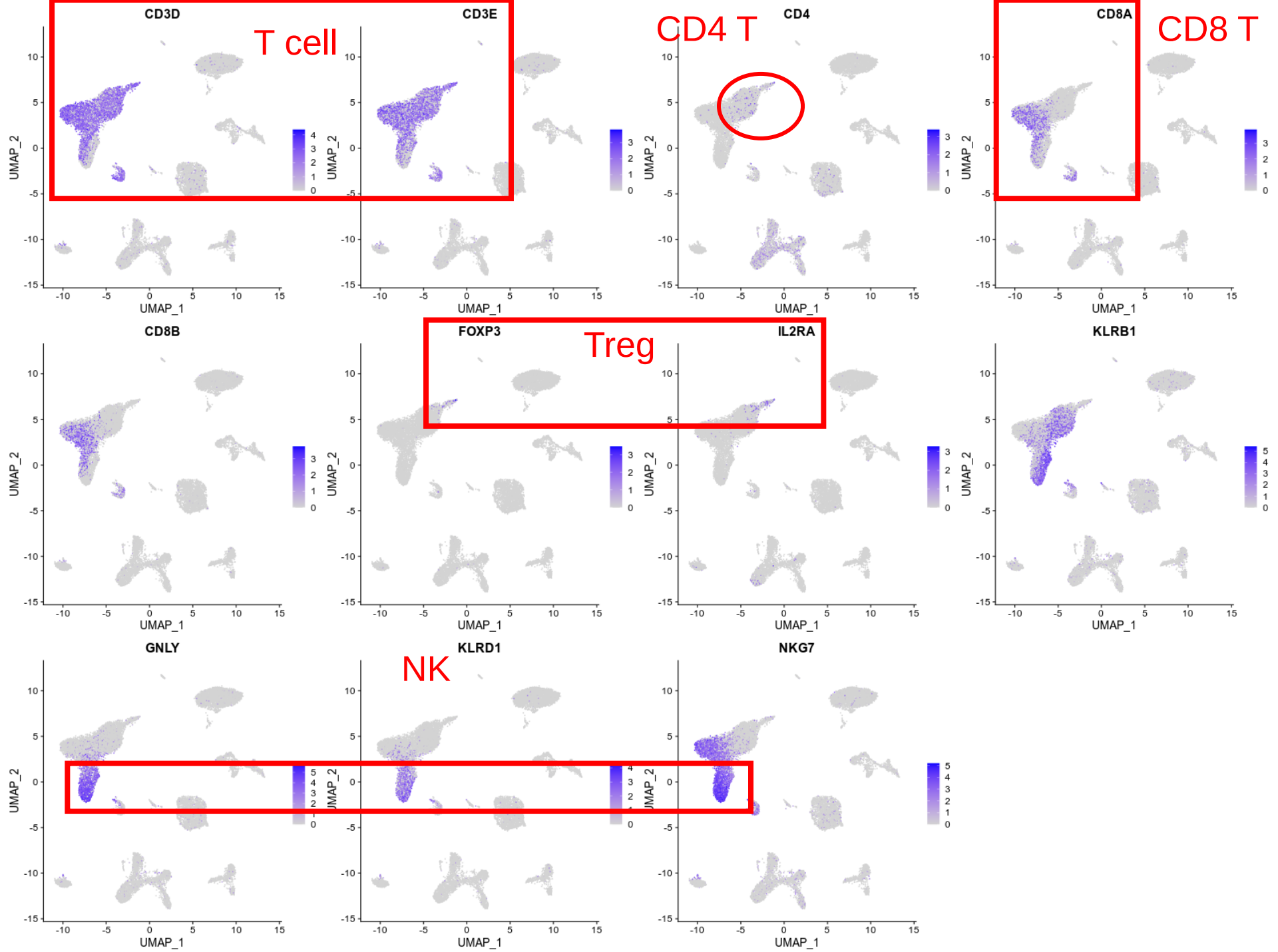
Epithelial cell: "SLC26A7", "EPCAM", "MUC1"

	p_val	avg_log2FC	pct.1	pct.2	p_val_adj	cluster	gene
CRYAB	0	4.405264	0.996	0.158	0	0	CRYAB
RBP4	0	3.934706	0.880	0.032	0	0	RBP4
UGT2B7	0	3.844966	0.950	0.039	0	0	UGT2B7
SPP1	0	3.729567	0.924	0.081	0	0	SPP1
WFDC2	0	3.711430	0.954	0.081	0	0	WFDC2
CLU	0	3.401655	0.915	0.089	0	0	CLU
DEFB1	0	3.202296	0.934	0.110	0	0	DEFB1
TNFRSF12A	0	2.974357	0.872	0.094	0	0	TNFRSF12A
PDZK1IP1	0	2.943278	0.761	0.028	0	0	PDZK1IP1
KRT18	0	2.863520	0.847	0.068	0	0	KRT18
TFPI2	0	2.857835	0.692	0.022	0	0	TFPI2
AC073218.2	0	2.825047	0.808	0.029	0	0	AC073218.2
SOSTDC1	0	2.822084	0.679	0.018	0	0	SOSTDC1
IGFBP6	0	2.698095	0.809	0.064	0	0	IGFBP6
CXCL14	0	2.682915	0.777	0.053	0	0	CXCL14
TMEM176A	0	2.603952	0.756	0.043	0	0	TMEM176A
SLPI	0	2.561452	0.519	0.020	0	0	SLPI
TSPAN1	0	2.555455	0.718	0.028	0	0	TSPAN1

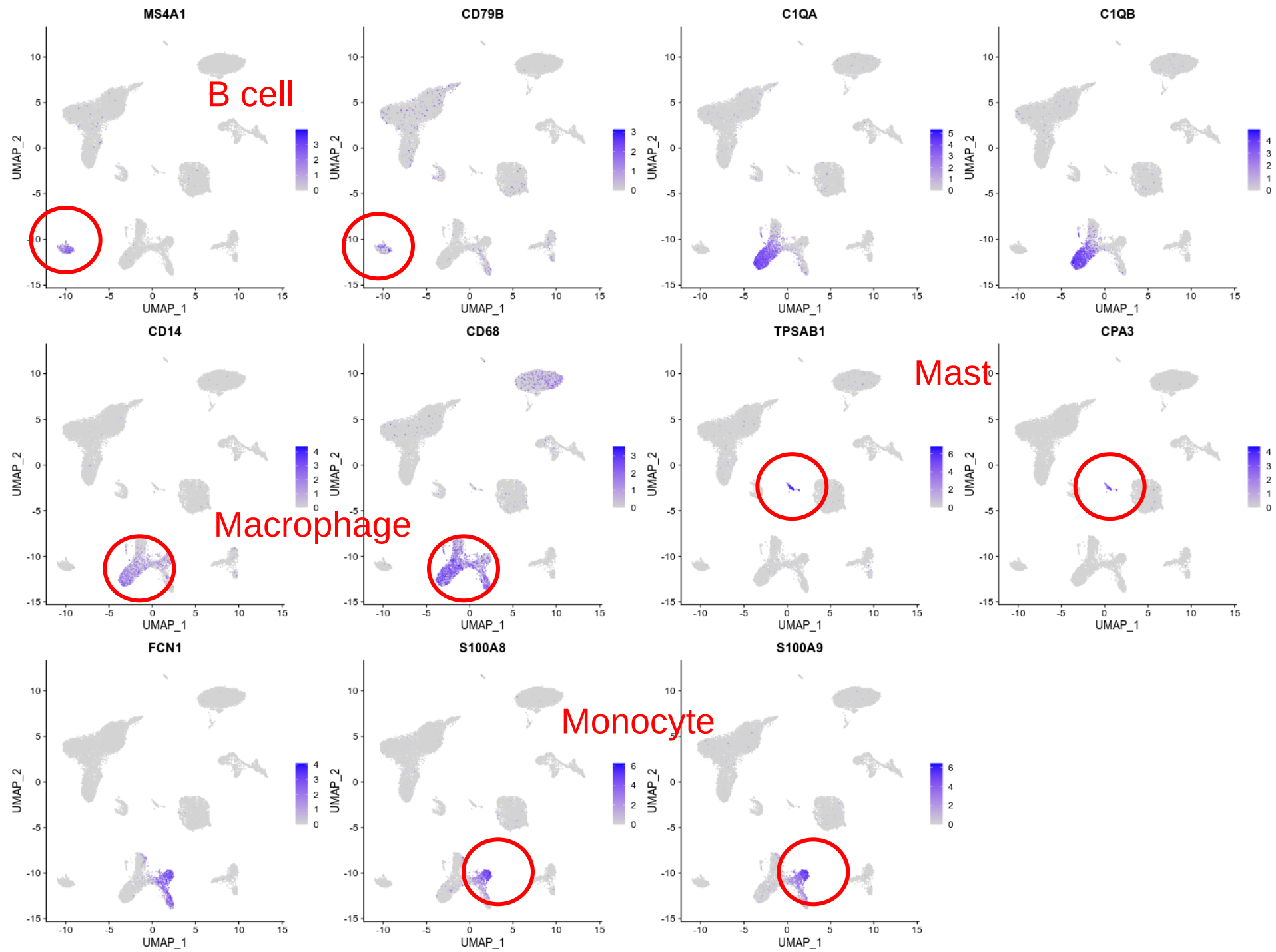
- Celltype annotation



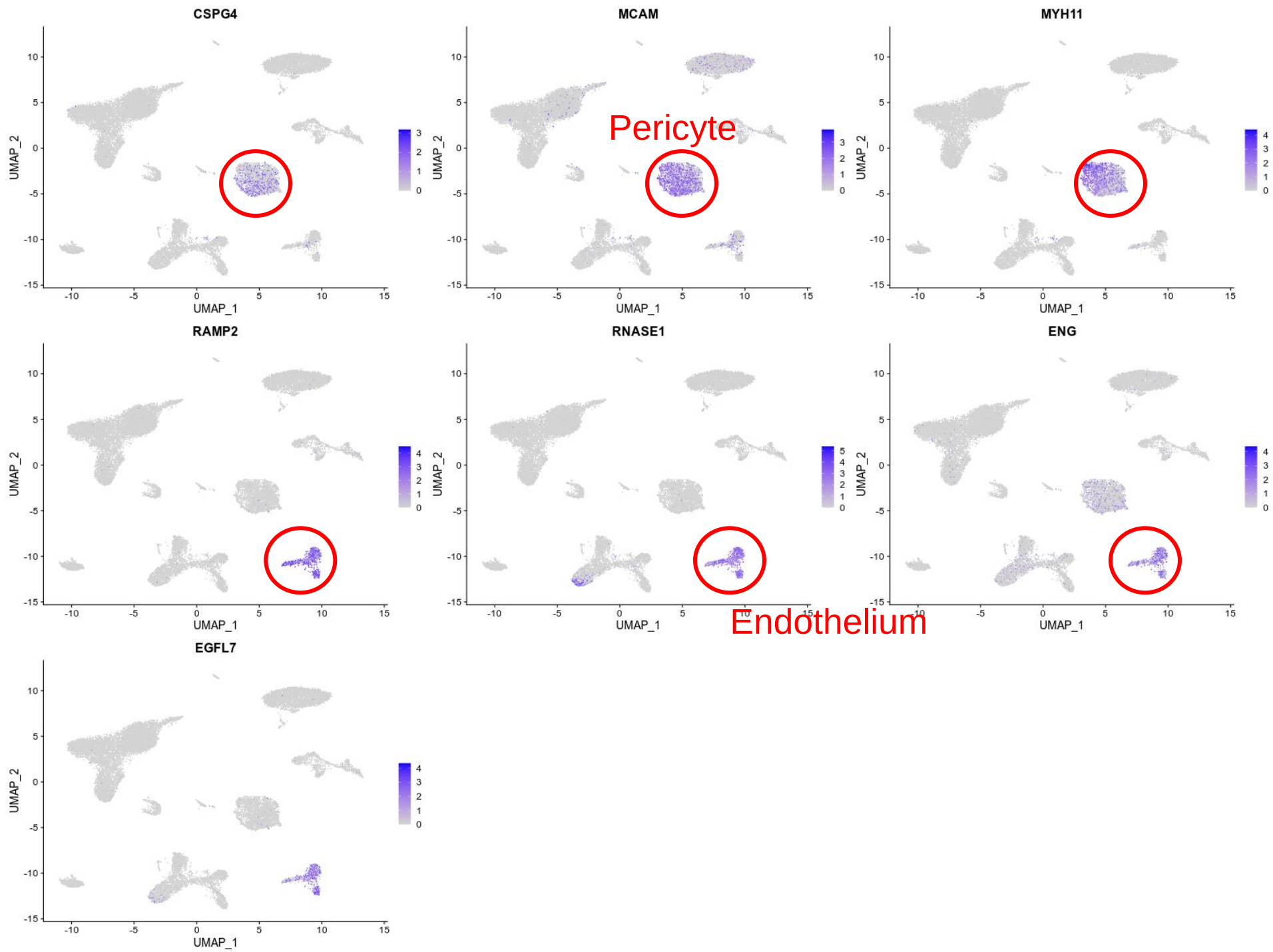
T cell and NK



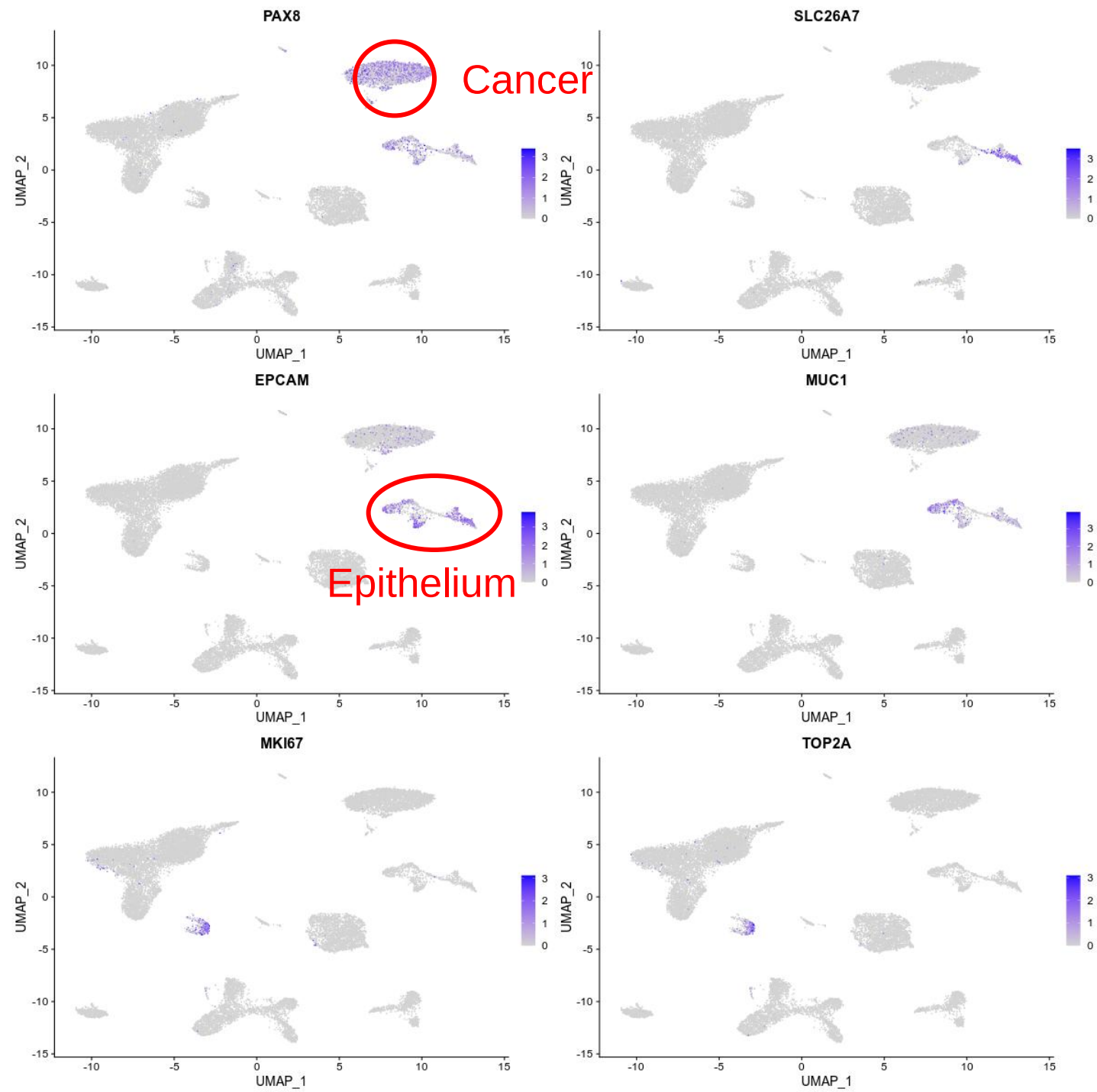
Bcell and myeloid



Stroma cell



Epithelial and tumor



• Differentially expressed gene analysis between two groups

- Wilcoxon rank sum test between two groups for each cell type (or cluster)
- Nonparametric approach (does not require a specific distribution of data)
- Adjusted p-value, average_Log2FC + expression cell ratio (pct.1, pct.2)



	p_val	avg_log2FC	pct.1	pct.2	p_val_adj
TMSB4X	1.180041e-123	0.8131724	1.000	1.000	3.863218e-119
CD74	1.336753e-110	2.4380974	0.841	0.460	4.376260e-106
HLA-DRA	6.658395e-84	2.6055562	0.616	0.121	2.179825e-79
RPS29	9.853846e-84	-0.6238238	0.997	1.000	3.225952e-79
RPS14	5.520321e-81	-0.6245685	0.993	1.000	1.807243e-76
RGS1	8.877664e-78	1.3781774	0.872	0.485	2.906370e-73
RPS27	2.936975e-74	-0.4452566	1.000	1.000	9.615067e-70
RPLP2	8.556250e-74	-0.5271463	0.998	1.000	2.801145e-69
DUSP4	1.420529e-72	2.1971333	0.553	0.075	4.650527e-68
RPL3	5.104196e-72	-0.6507585	0.982	0.992	1.671012e-67
EEF1A1	3.198898e-69	-0.5291642	0.997	1.000	1.047255e-64
RPS3	7.501267e-69	-0.5543631	0.994	0.998	2.455765e-64
HLA-DRB1	3.762663e-62	1.7883327	0.662	0.273	1.231821e-57
HLA-DPB1	1.903542e-61	1.7512534	0.649	0.275	6.231815e-57

GZMA: Tumor_CD8 T cell > Normal_CD8 T cell

→ Cytotoxic

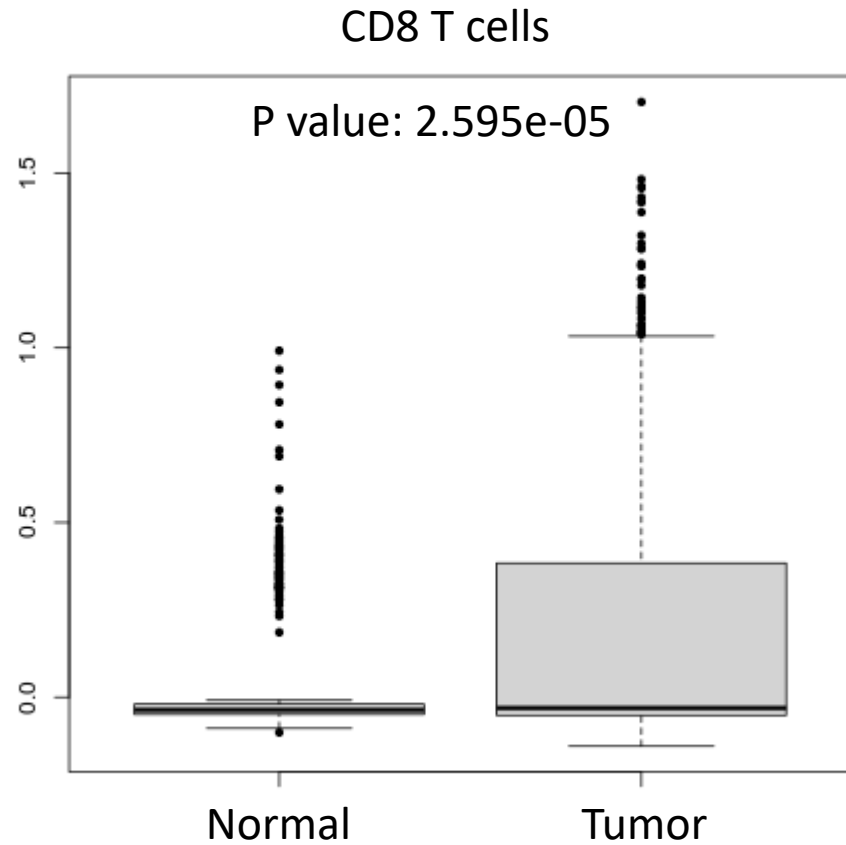
HLA-DRA, HLA-DRB1, HLA-DPB1: Tumor_CD8 T cell > Normal_CD8 T cell

→ activated

- # Signature analysis

Tumor infiltrating T cell → might be exhausted

Exhaustion signature: PDCD1, CTLA4, HAVCR2, LAG3, TOX



→ T cells in the tumor-microenvironment are exhausted

